

Open Science and Reproducible Research

Part 2: reproducible data analysis

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Reproducibility and Replicability

- They have different meanings in different disciplines.
- In computer science/medicine/epidemiology:
 - **Reproducibility** refers to instances in which the *original researcher's data* and *computer codes* are used to regenerate the results
 - **Replicability** refers to instances in which a researcher collects new data to arrive at the same scientific findings as a previous study.

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Part 1: GitHub Intro



What is version control?

- A system that records changes in files
- Everybody has used it, even if it's a system you have made yourself!



1 Thesis
Semi-Final



2 Thesis Final



3 Thesis Super
Final



4 Thesis Super
dooper Final



5 Thesis Super
dooper final
najud ni



6THESIS FINAL
NA TALAGA



7THESIS Final
najud Salamat
Lord



8 AMBOT NGA
THESIS



9 LORD GUSTO
KO MO
GRADUATE
HUHUUH

Why use version control?

- Backup/restore/saving
- Synchronization
- Track changes (and ownership)
- Seeing your progress
- Professional way to create code

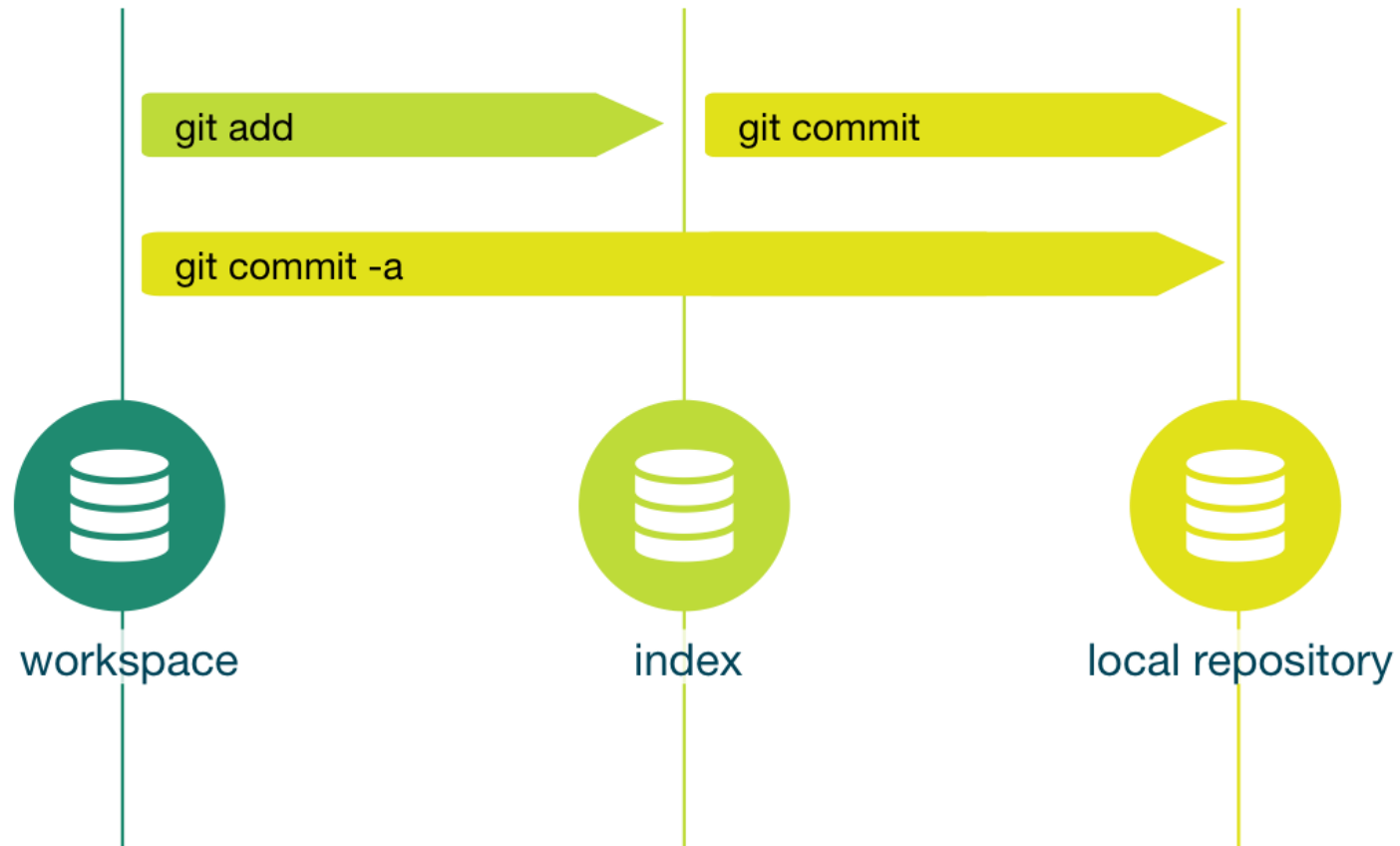


Why use version control?

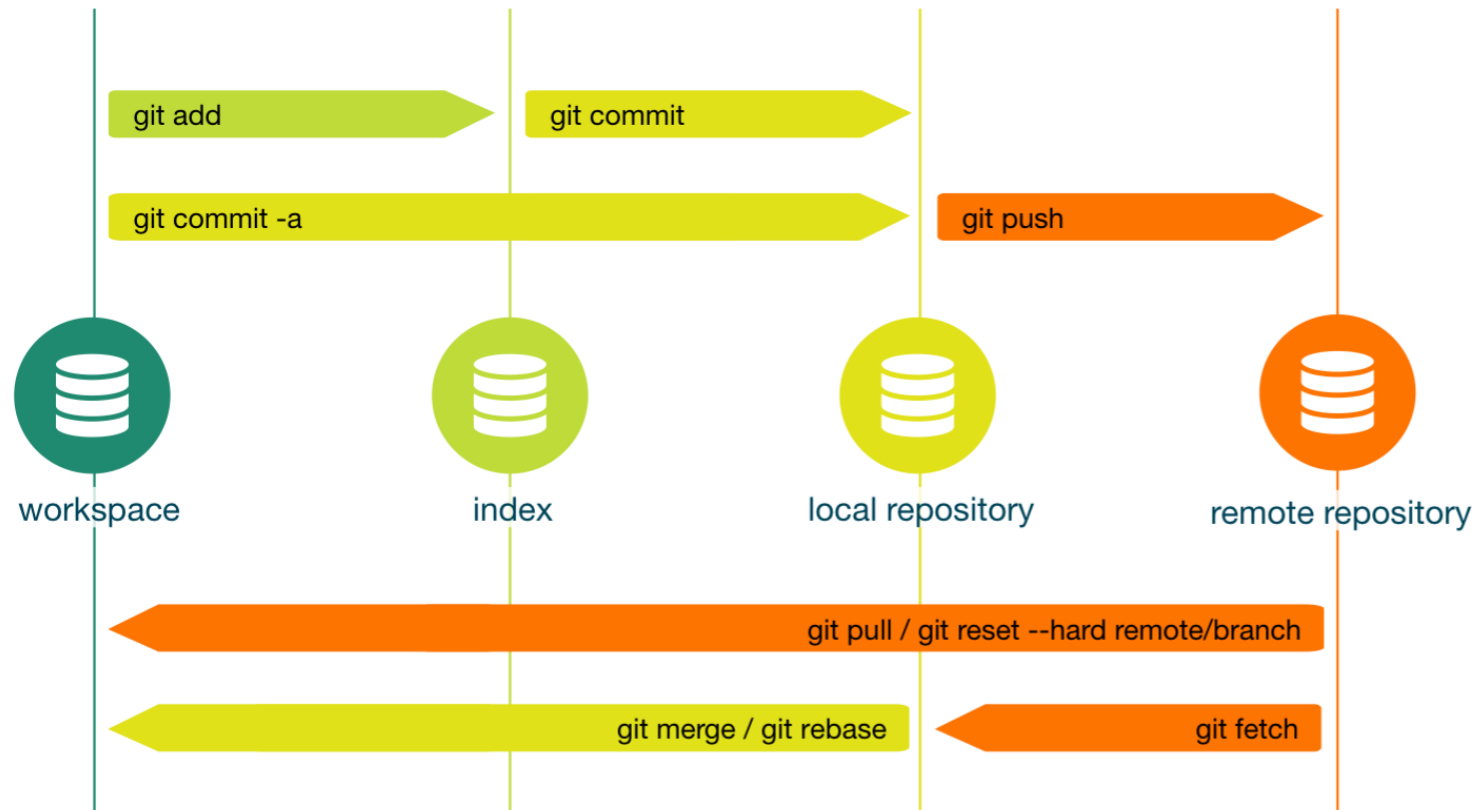
- Backup/restore/saving
- Synchronization
- Track changes (and ownership)
- Seeing your progress
- Professional way to create code



Git is the most common version control system



Git paradigm



GitHub: an online repository host

- Why GitHub?
- Public and private repositories
- Tracking of your work
- Easy collaboration



Some key dos and don'ts

Do:

- Commit often
- Commit small changes as you go
- Write useful commit messages

Don't:

- Commit untested changes
 - Add data files
 - Add un-trackable files
 - Add enormous files
- 

Best practices

- **.gitignore**
 - **Avoid uploading original data.**
- Add a README
- Add a License
- Do not put a repo in Dropbox/Google Drive...

Practical: account

1. Create a GitHub account at: <https://github.com/join>
2. Use your NTU address to get a free academic account.
3. When logged in, find the green “New” button to create a new repository
4. Give it a name: “*test_repo*” and a description.
5. Make it public
6. Press create.

Practical: Setup

- Install GitHub Desktop
 - Go to: <https://docs.github.com/en/desktop/installing-and-authenticating-to-github-desktop/installing-github-desktop> and run the download.
(Or Google GitHub Desktop)
 - Install and accept all default options.



[Overview](#) [Release Notes](#) [Help](#)

GitHub Desktop

Focus on what matters instead of fighting with Git. Whether you're new to Git or a seasoned user, GitHub Desktop simplifies your development workflow.

[Download for Windows \(64bit\)](#)

✦ **Feeling brave?**

Try new features in the  [Beta Channel](#) before they're released.

Prefer the MSI?

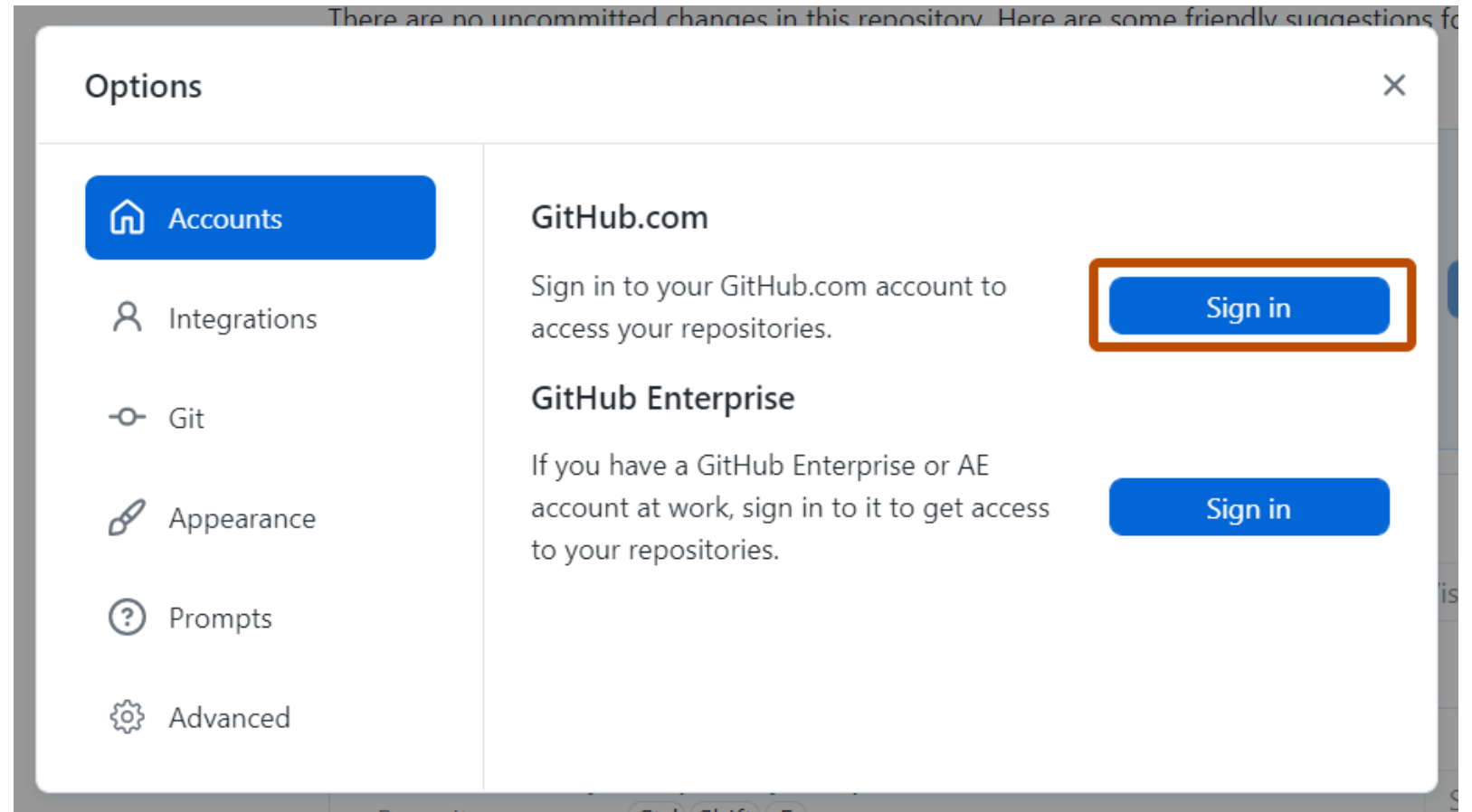
[Download for Windows \(MSI\)](#)

macOS?

[Download for macOS](#)

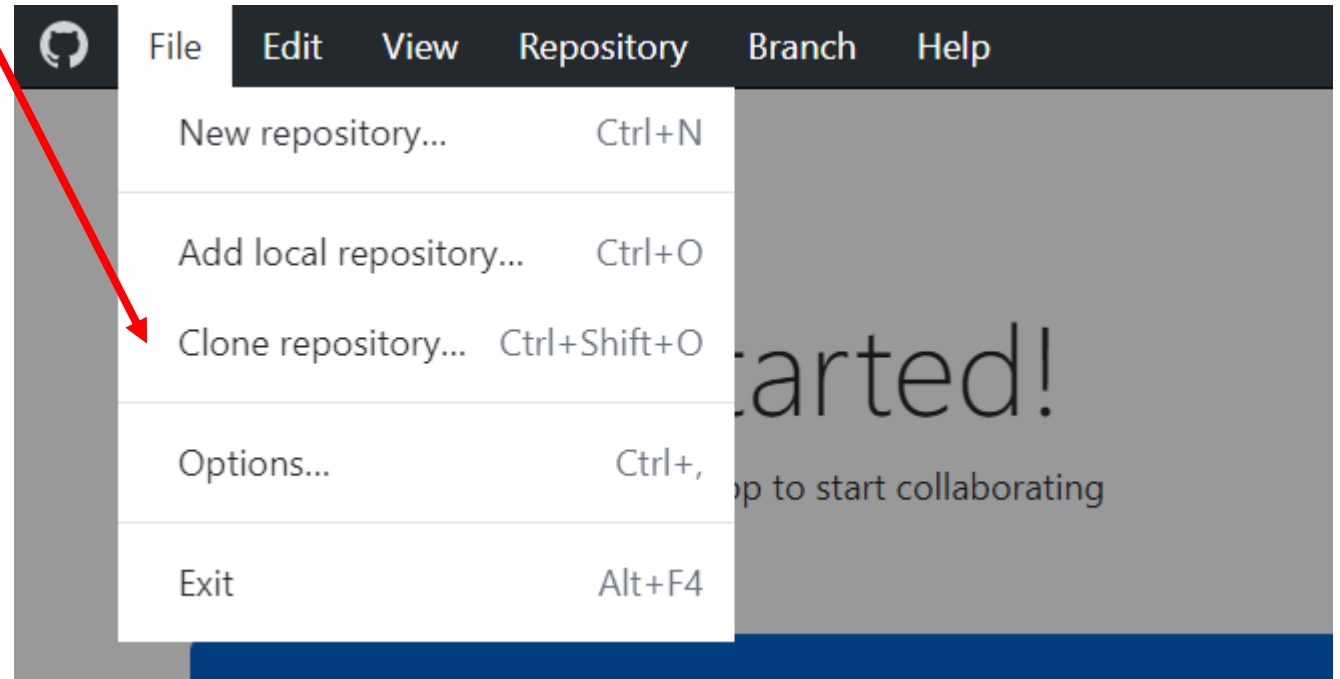
Using GitHub Desktop

- Sign in your account



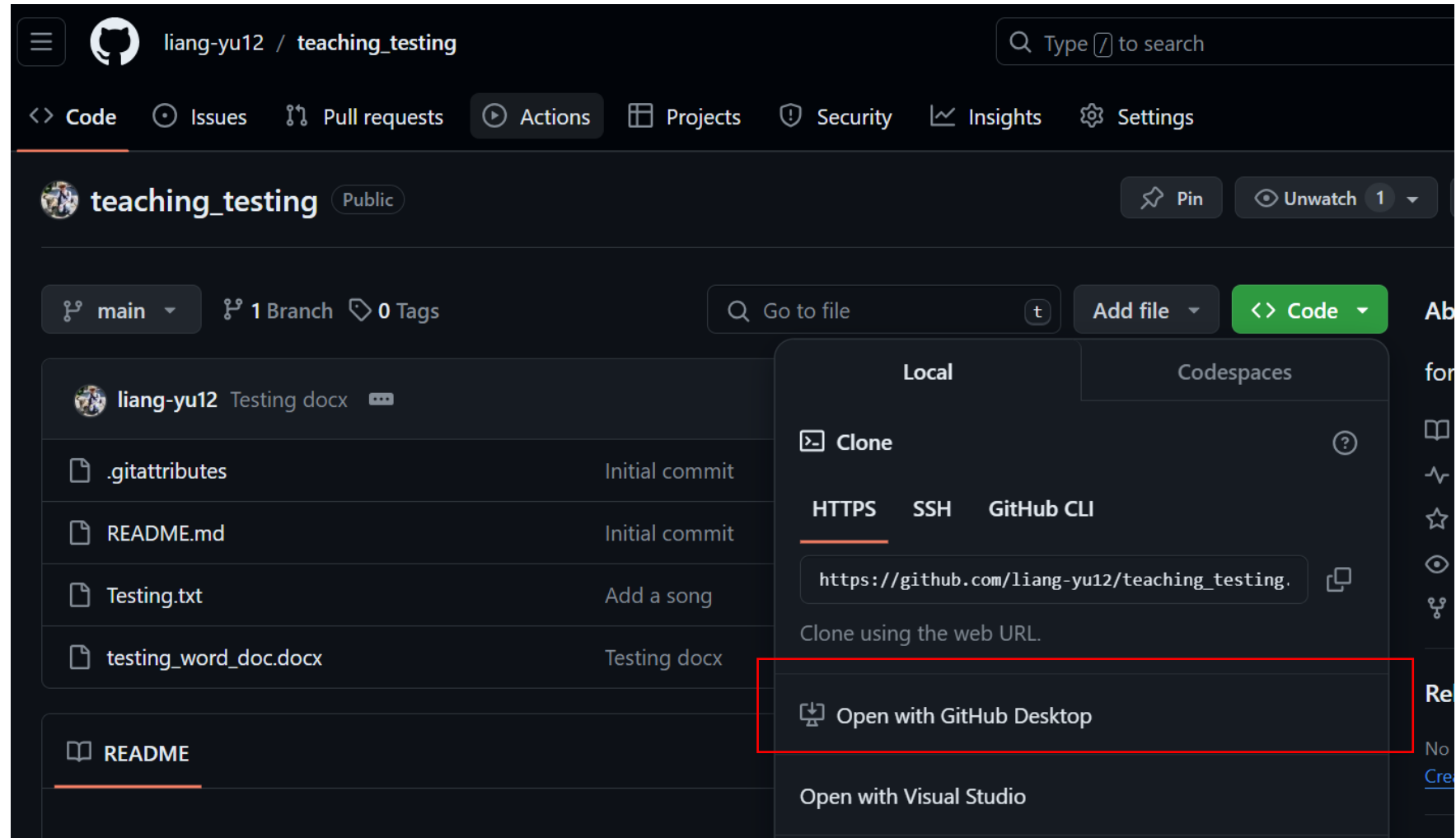
Clone the repository

- It's just like creating a Google Drive/Dropbox folder on your computer
- You can also clone an existing folder from your online account



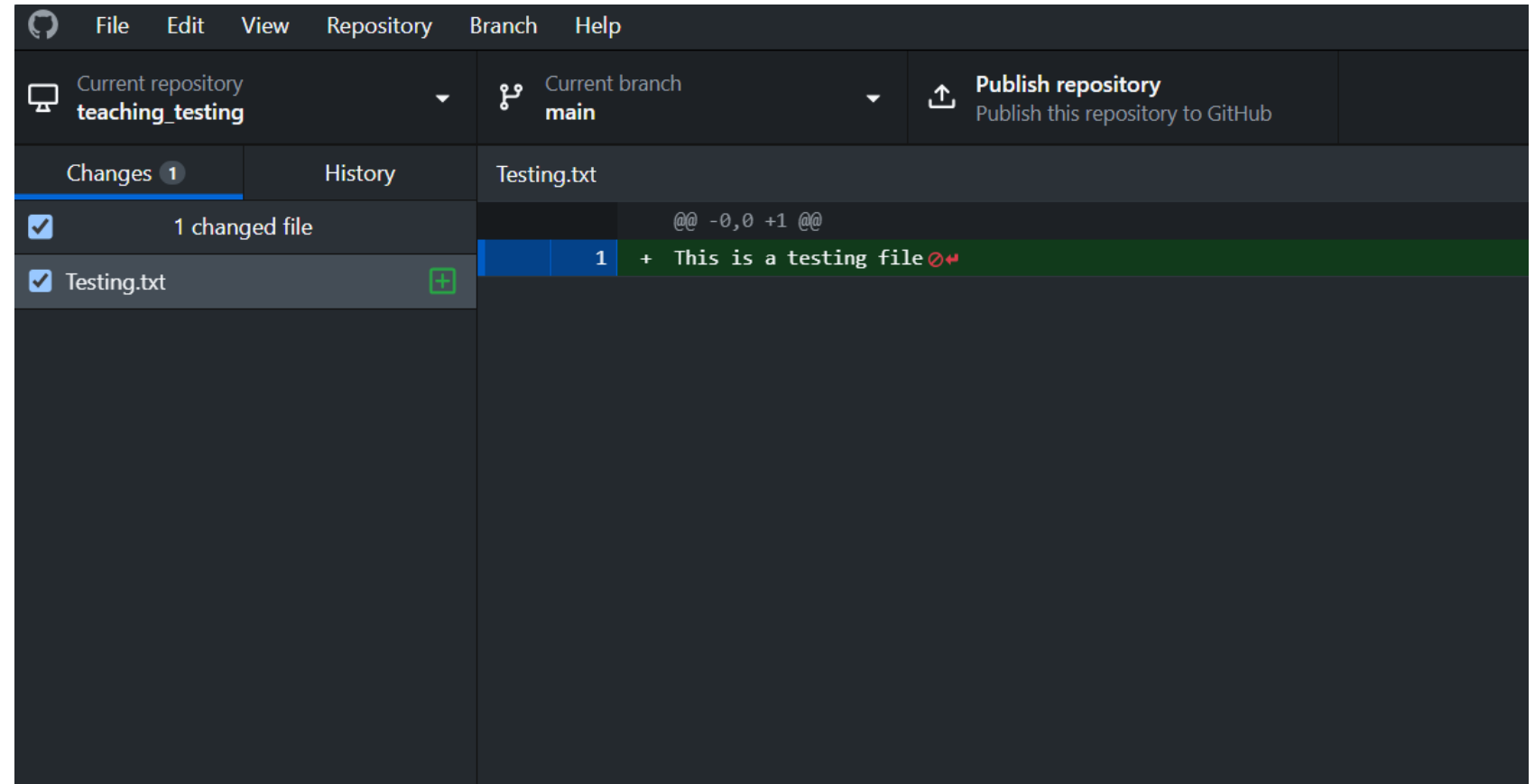
Another way to clone the folder

- Then choose a folder for your repo
- Do not put a repo in Dropbox/Google Drive



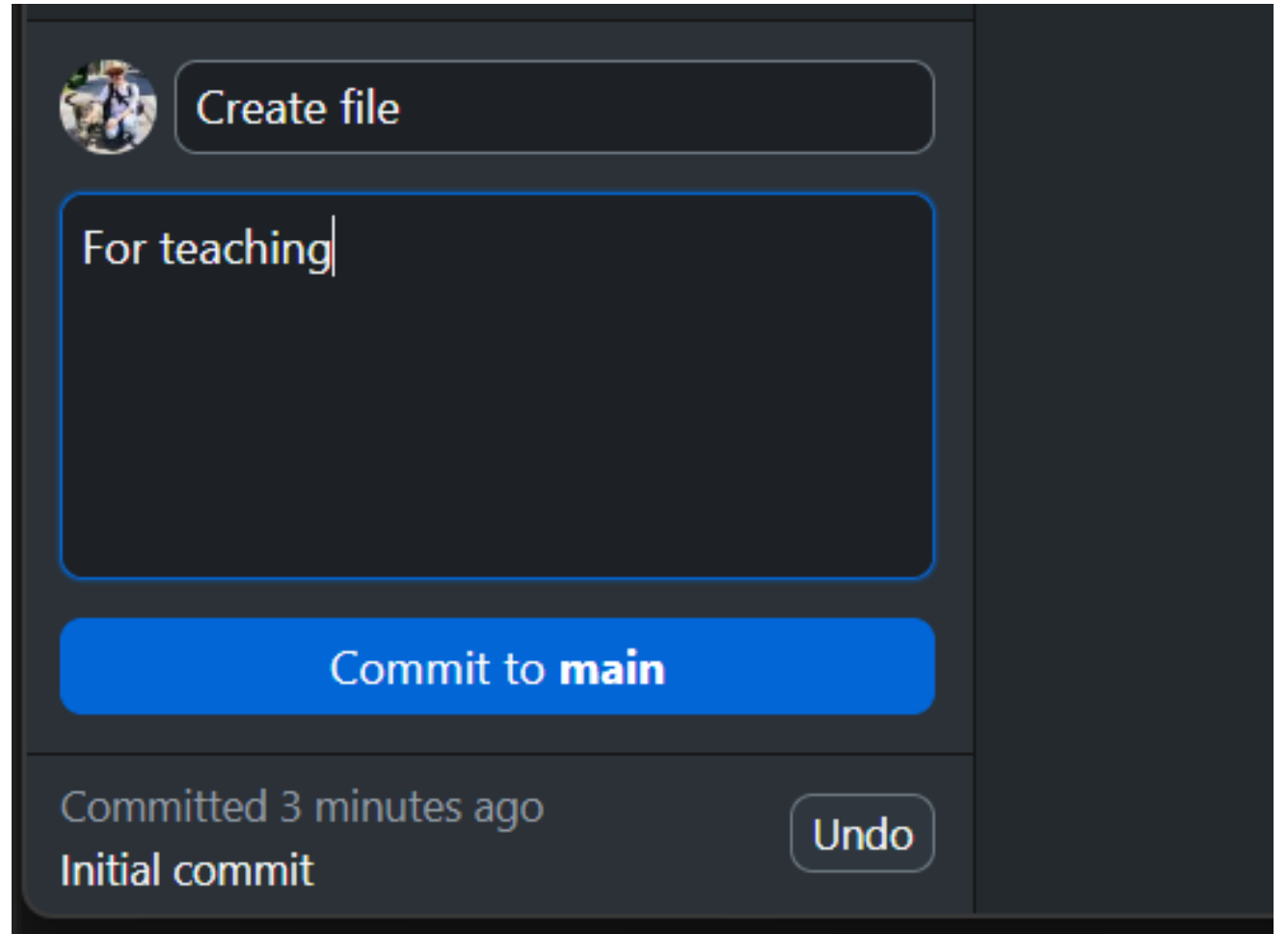
Open the folder on your laptop and create a file

- Create a text file in the folder
- Meanwhile on the GitHub Desktop, you'll see your new file



Commit your file

- Write down what you had done to your file
- Keep it short and clear
- Then click Commit to main

A screenshot of a GitHub commit interface. At the top left is a circular profile picture of a person. To its right is a button labeled "Create file". Below this is a large text input area containing the text "For teaching". At the bottom of the input area is a prominent blue button labeled "Commit to main". Below the button, the text "Committed 3 minutes ago" is displayed, followed by "Initial commit" in a slightly larger font. To the right of this text is a button labeled "Undo".

Create file

For teaching

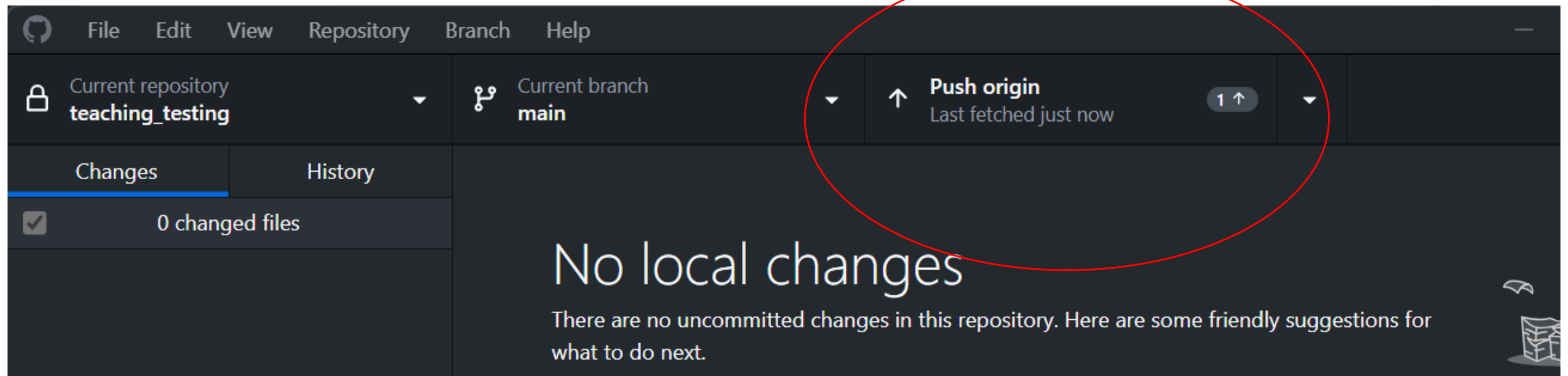
Commit to **main**

Committed 3 minutes ago
Initial commit

Undo

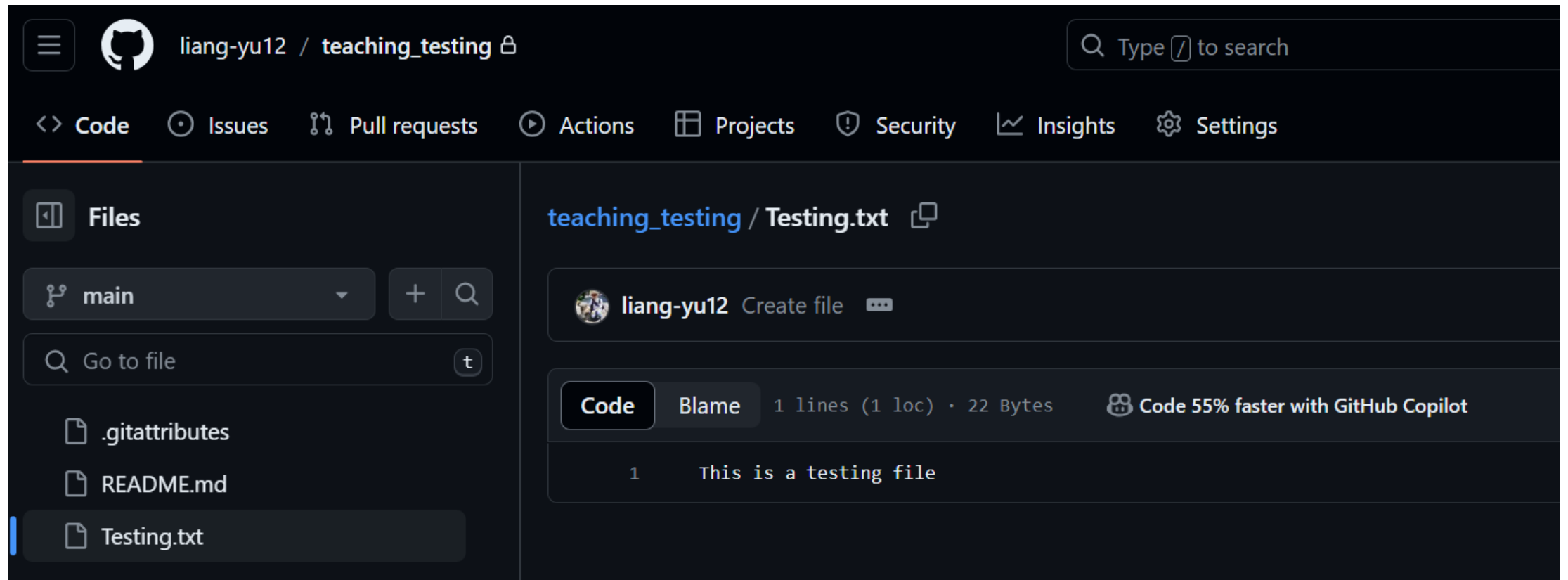
Push it!

- Sync your changes online



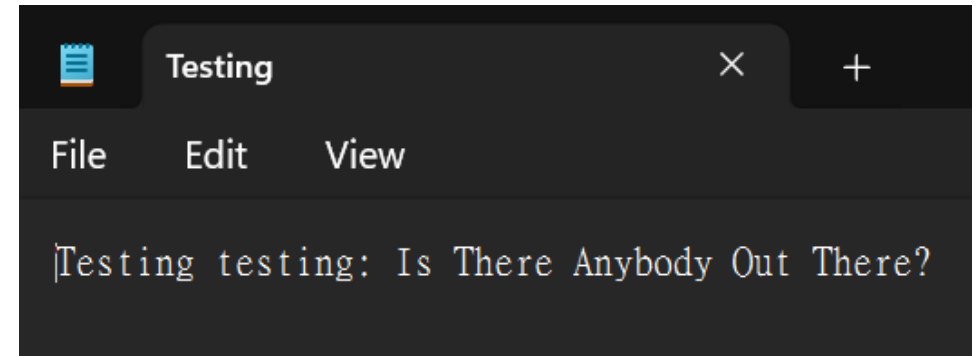
Now check your online repo

- The text file has been uploaded.

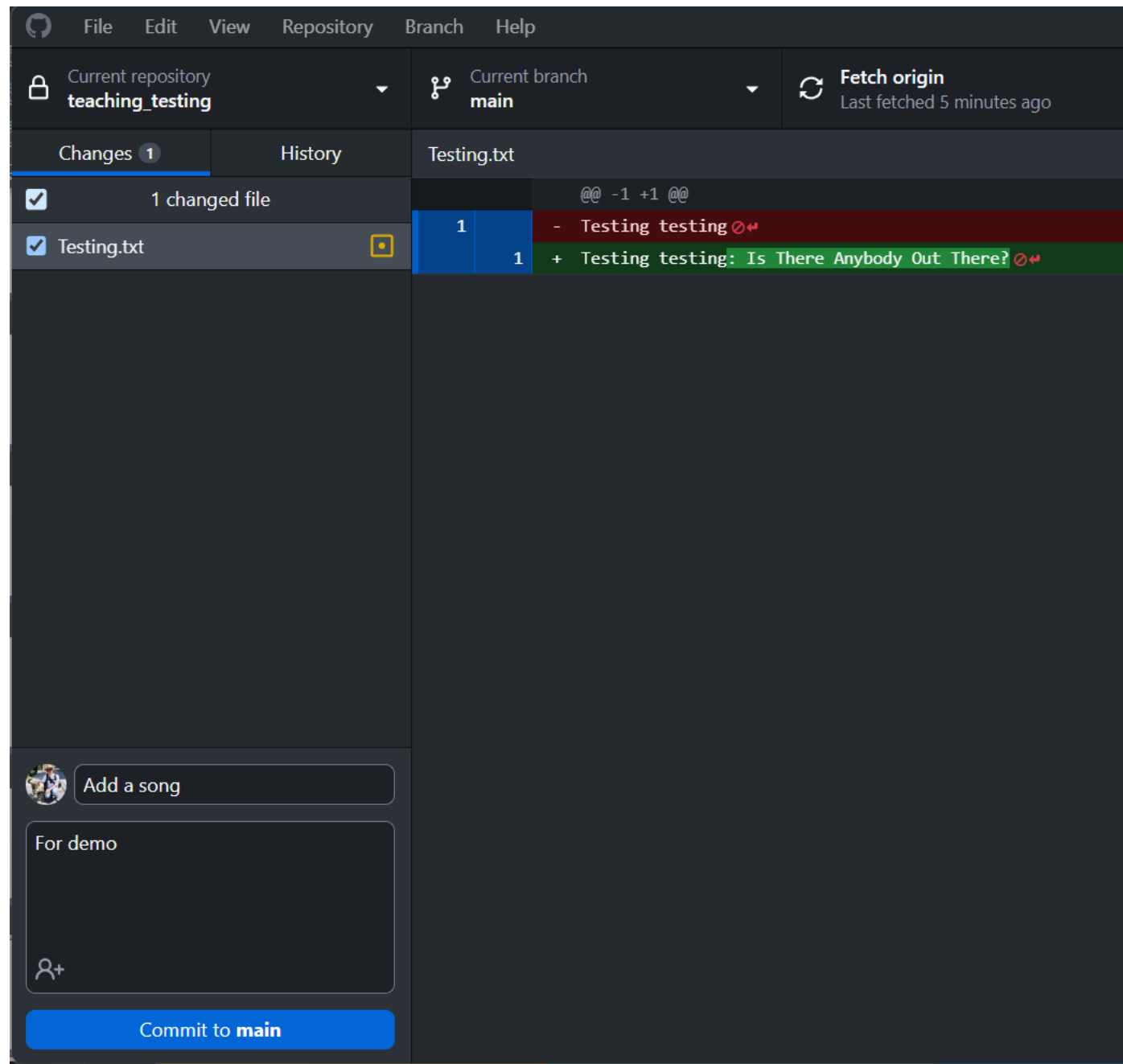


Edit your text file

- Add some words and save the file.
- Then check your GitHub Desktop



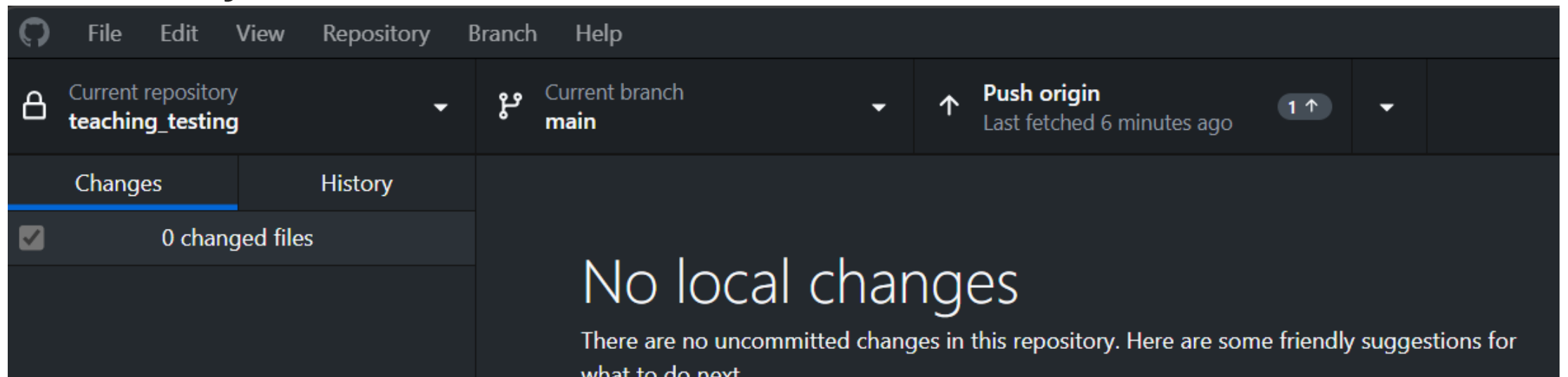
**Add another
commit:**
explain the
changes



**Changes you
made**

Push it!

- Push your commit online



- Now move to your online GitHub page

Look at your commits

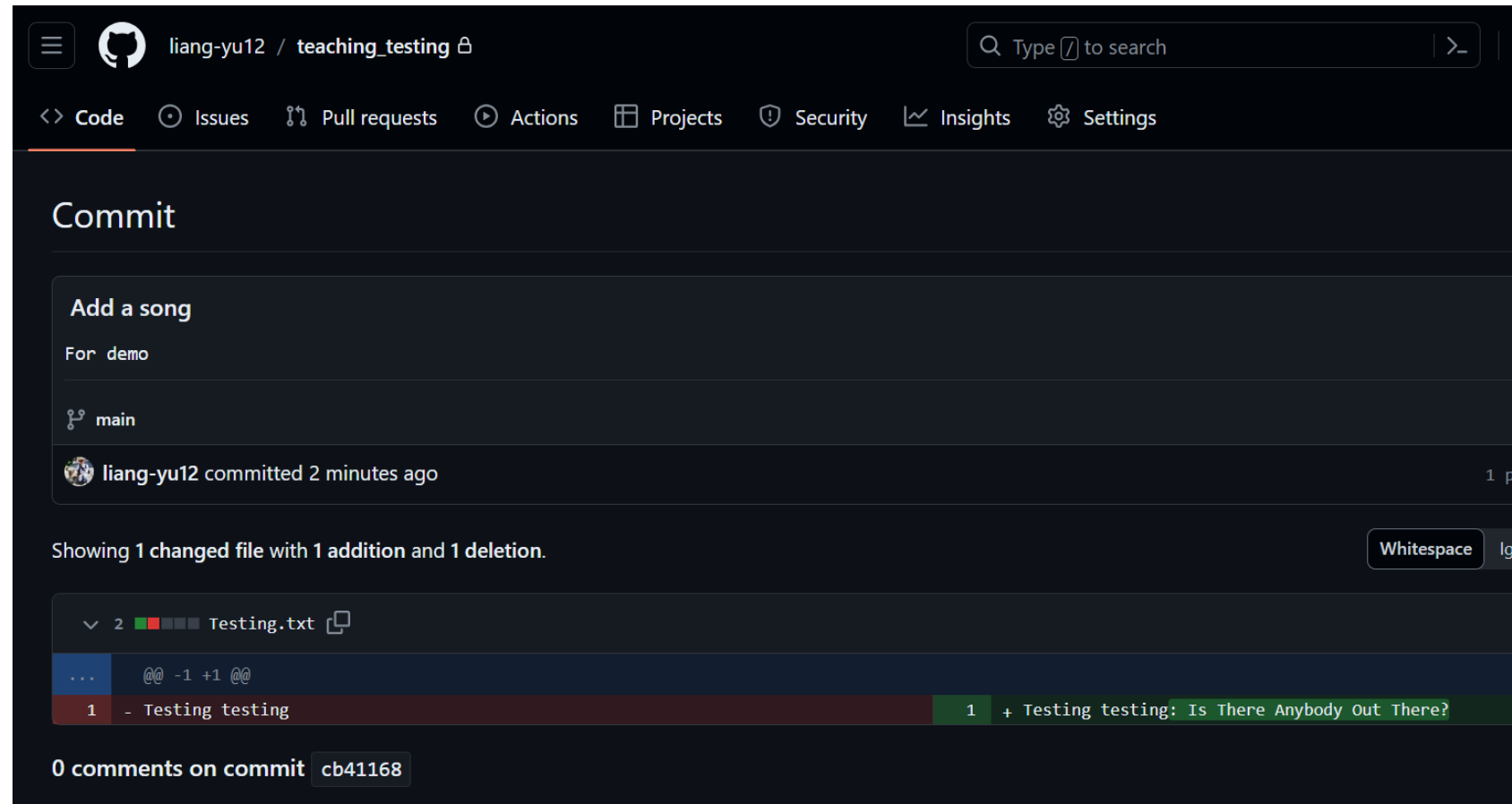
The screenshot shows the GitHub interface for a repository named 'teaching_testing'. The repository is private and has 1 branch (main) and 0 tags. The commit history table is as follows:

Commit Message	Commit Hash	Time Ago
Initial commit	cb41168	16 minutes ago
Initial commit		16 minutes ago
Add a song		1 minute ago

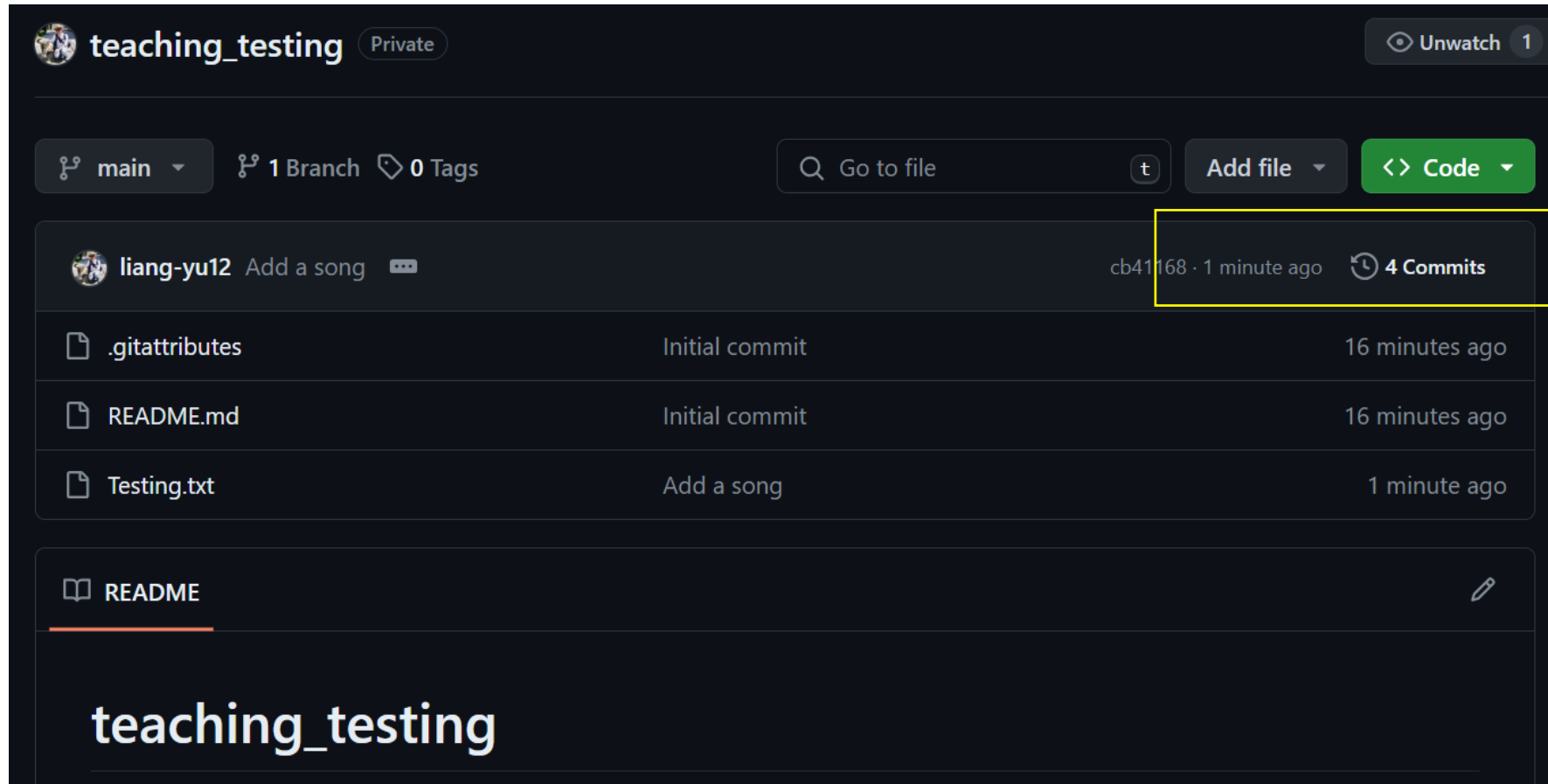
The commit history is displayed in a table with three columns: Commit Message, Commit Hash, and Time Ago. The commit messages are 'Initial commit', 'Initial commit', and 'Add a song'. The commit hashes are 'cb41168' and an empty string. The times are '16 minutes ago', '16 minutes ago', and '1 minute ago'. A yellow box highlights the commit messages.

Commit details

- On the page, you'll see what has been changed, and the reason why you changed it



All commits



The screenshot shows the GitHub interface for a repository named 'teaching_testing'. The repository is marked as 'Private'. At the top, there are navigation elements including a dropdown for the 'main' branch, indicating '1 Branch' and '0 Tags'. A search bar labeled 'Go to file' is present, along with buttons for 'Add file' and 'Code'. The commit history section shows a commit by user 'liang-yu12' with the message 'Add a song', committed '1 minute ago' with commit hash 'cb41168'. This commit has '4 Commits'. Below this, a list of files is shown: '.gitattributes' (Initial commit, 16 minutes ago), 'README.md' (Initial commit, 16 minutes ago), and 'Testing.txt' (Add a song, 1 minute ago). At the bottom, the 'README' file is selected, showing the repository name 'teaching_testing'.

teaching_testing Private Unwatch 1

main 1 Branch 0 Tags Go to file Add file <> Code

liang-yu12 Add a song cb41168 · 1 minute ago 4 Commits

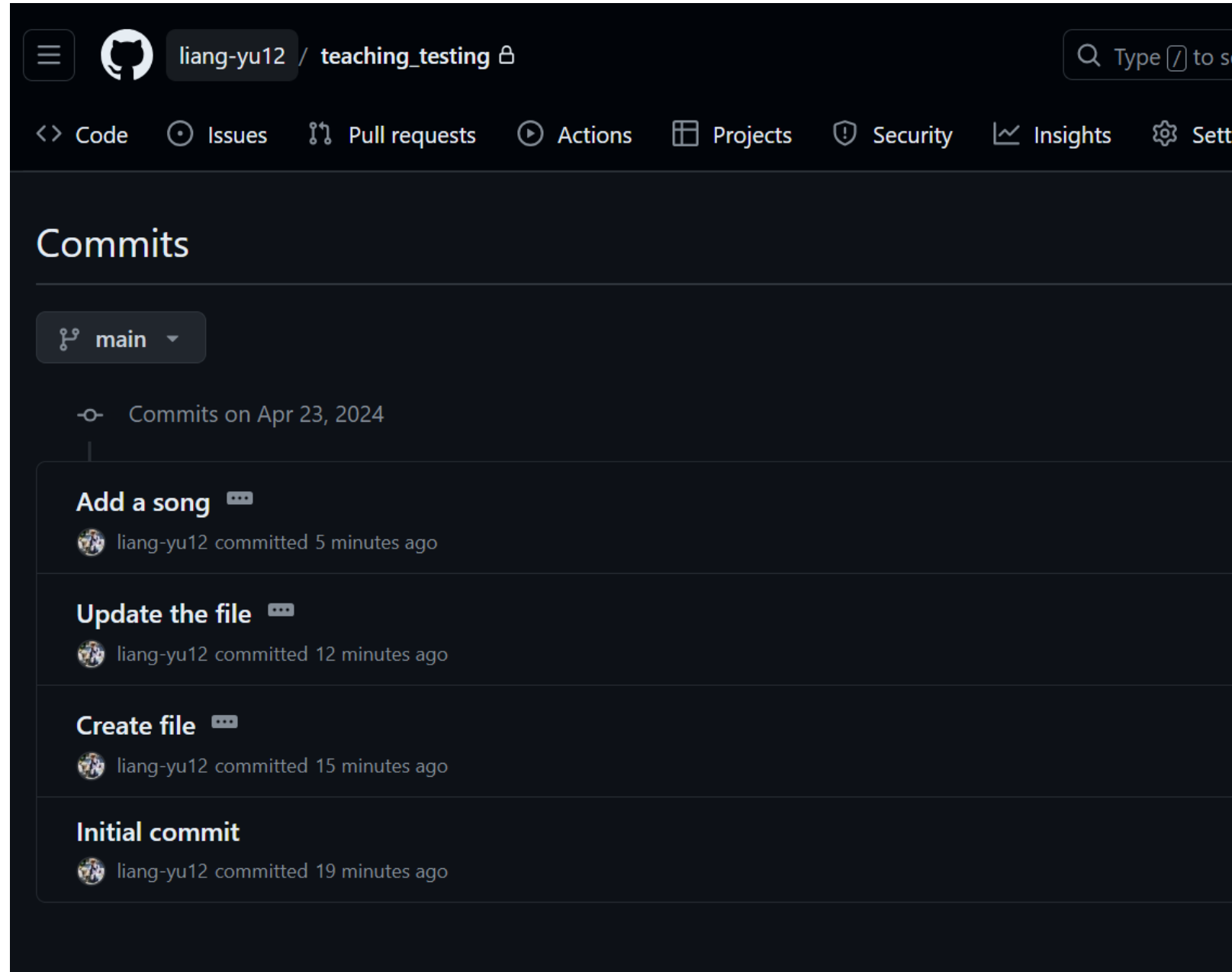
.gitattributes	Initial commit	16 minutes ago
README.md	Initial commit	16 minutes ago
Testing.txt	Add a song	1 minute ago

README

teaching_testing

Commit history

- Every change has been recorded.
- You can see the history

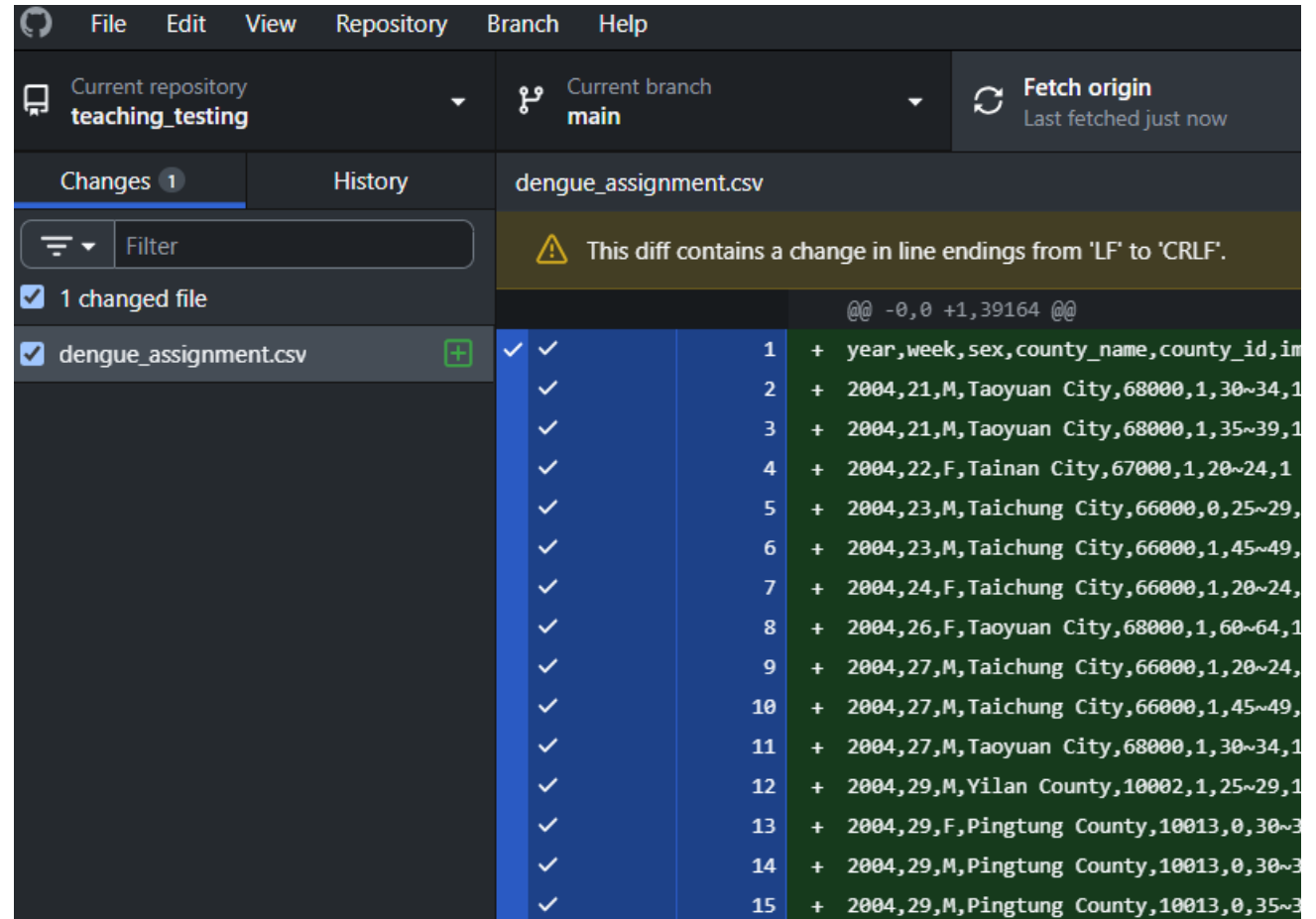


Ignore data files

- Important: Do **not** upload your dataset to GitHub.
 - We need to ignore them by setting up the *.gitignore* file

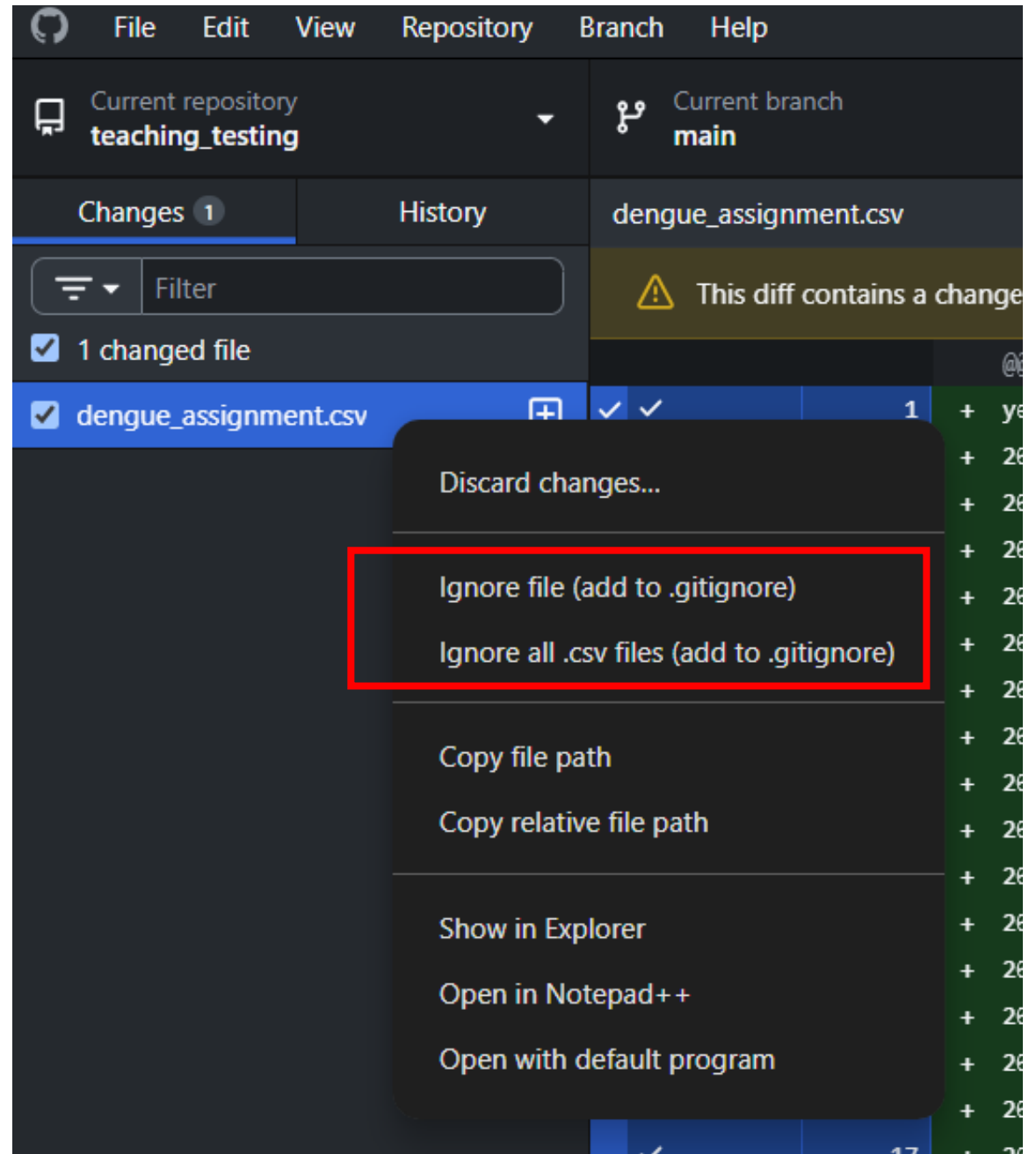
Ignore dataset files

- Example: if we paste a dataset.csv file to the local GitHub folder



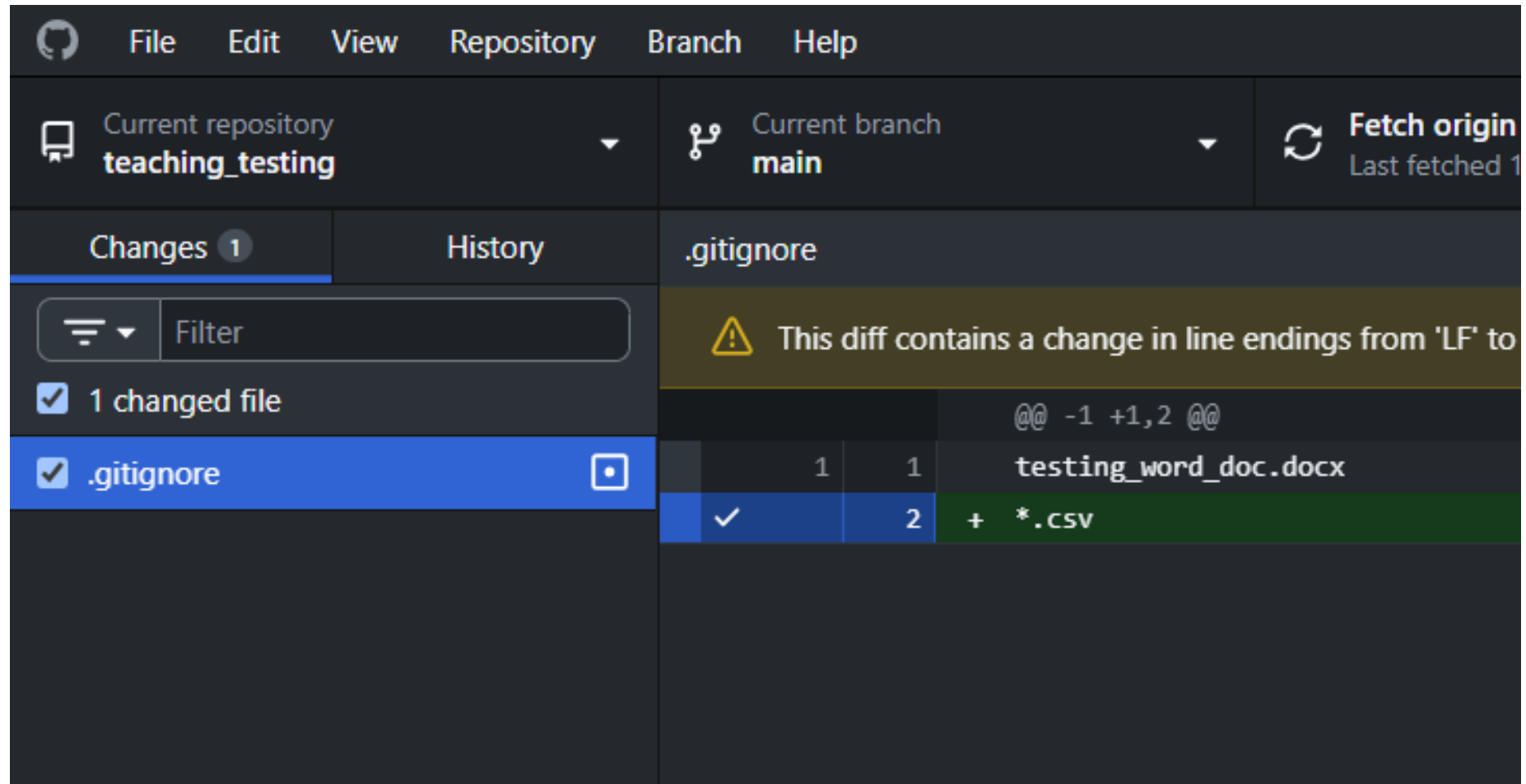
Ignore dataset files

- Right-click the file, and choose ignore file
 - Suggest ignoring all .csv files
 - It will add a line to the *.gitignore* file



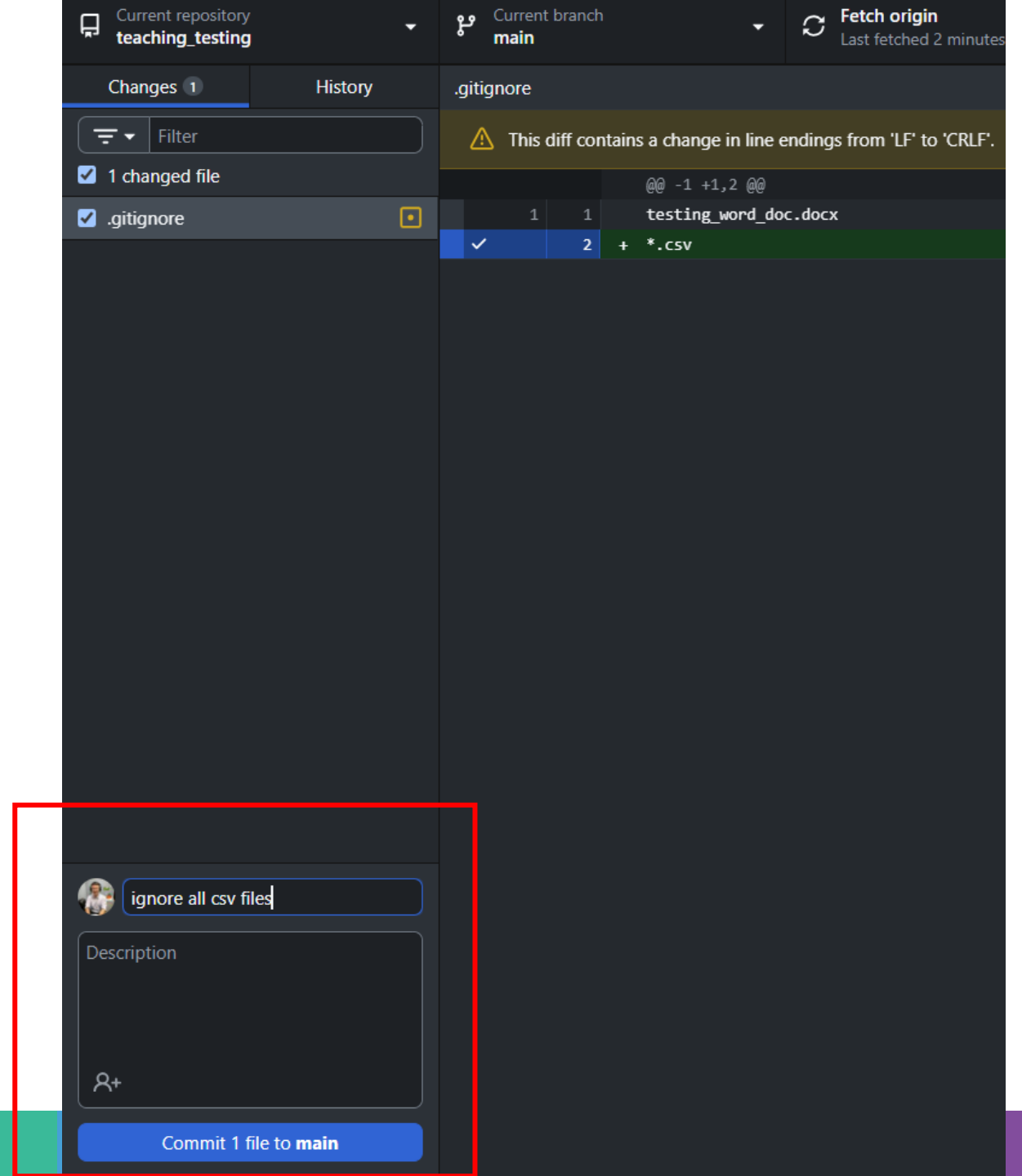
Ignore dataset files

- *.csv file will be added to .gitignore



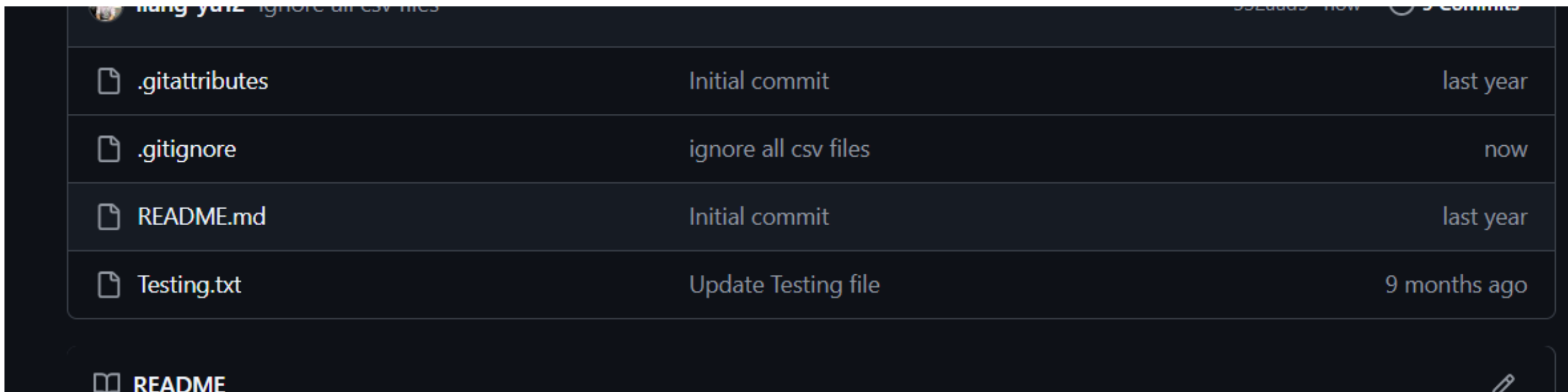
Ignore dataset files

- We still need to commit the changes to the *.gitignore* file.
- And push it



Ignore dataset files

- Check the remote GitHub
 - No csv file was uploaded.



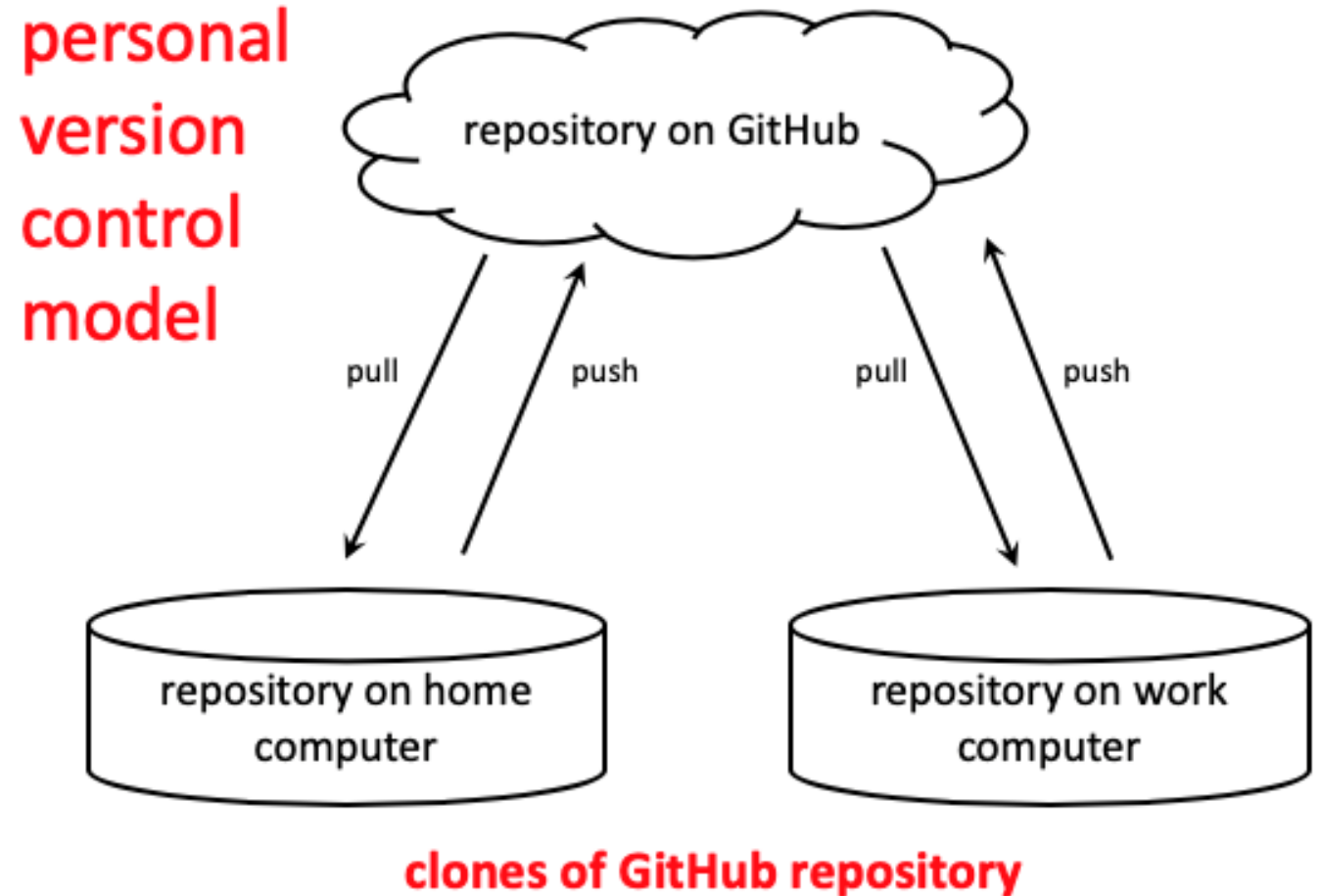
A screenshot of a GitHub repository's file list. The repository name is 'hang yu12' and the commit message is 'ignore all csv files'. The file list includes:

File	Commit Message	Time
 .gitattributes	Initial commit	last year
 .gitignore	ignore all csv files	now
 README.md	Initial commit	last year
 Testing.txt	Update Testing file	9 months ago

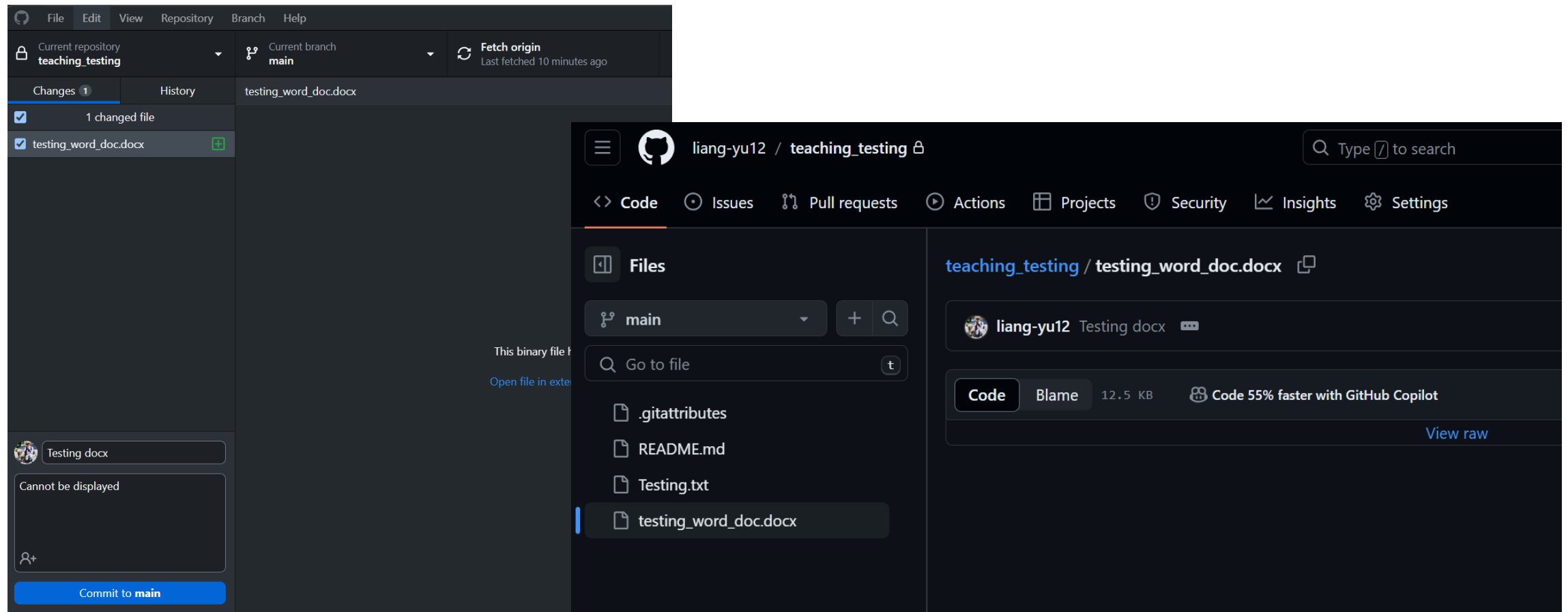
At the bottom, there is a 'README' link with a book icon and an edit icon.

Working on another machine

- You can pull the codes from your GitHub repo to another machine.
 - You can work on more than one laptop/PC
 - Don't need to worry about a stolen laptop anymore.

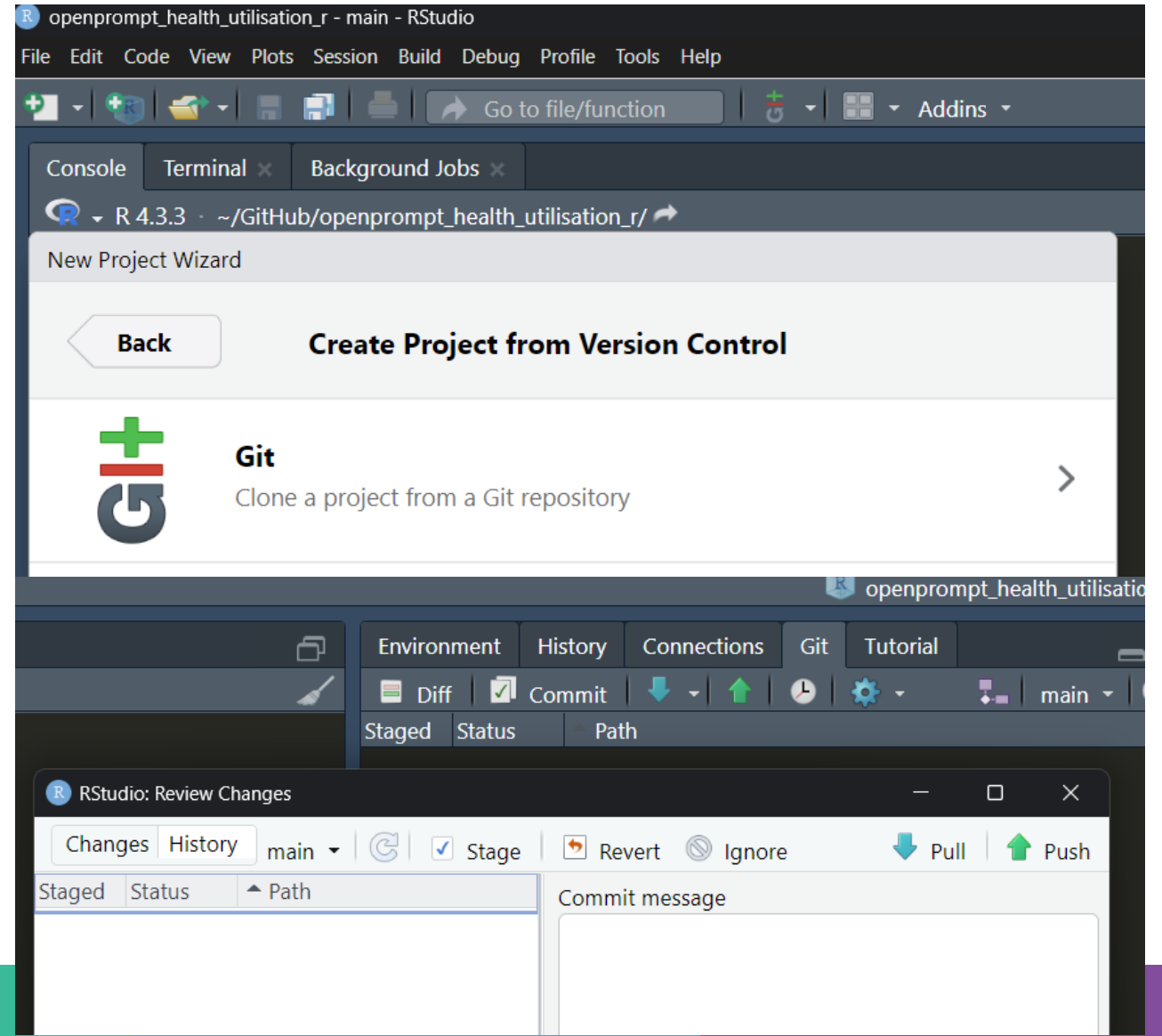


Sorry: GitHub is not good for MS Word



Bonus: using GitHub in RStudio

- For R users
- You can create an R project with Git integration using RStudio



Take home message

- Open Science: make scientific research more transparent, reproducible, and accessible. It could potentially reduce scientific fraud.
- GitHub is a great platform for code sharing and version control. Its commit function can make your research more transparent.



Thank You!

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